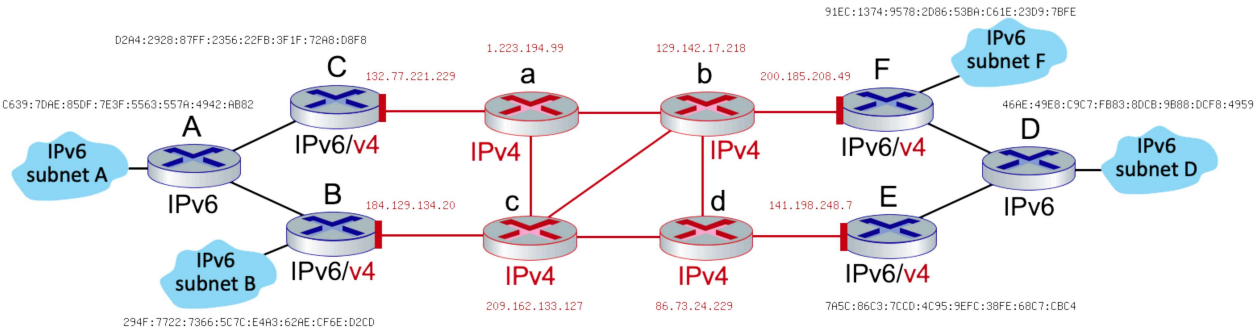
INTERACTIVE END-OF-CHAPTER EXERCISES

IPV6 TUNNELING AND ENCAPSULATION

Consider the network shown below which contains four IPv6 subnets, connected by a mix of IPv6-only routers(shaded blue), IPv4-only routers (shaded red) and dual-capable IPv6/IPv4 routers (shaded blue with red interfaces to IPv4 routers). You might want to review Section 4.3.4 in the textbook before doing this problem.



Suppose that a host of subnet D wants to send an IPv6 datagram to a host on subnet A. Assume that the forwarding between these two hosts goes along the path: D --> F --> b --> a --> c --> B --> A

QUESTION LIST

1. Is the datagram being forwarded from D to F an IPv4 or IPv6 datagram?

2. What is the source address of this D to F datagram?

3. What is the destination address of this D to F datagram?

4. Is this D to F datagram encapsulating another datagram? Yes or No.

5. Is the datagram being forwarded from F to b an IPv4 or IPv6 datagram?

6. What is the source address of this F to b datagram?

7. What is the destination address of this F to b datagram?

8. Is this F to b datagram encapsulating another datagram? Yes or No.

9. What is the source address of this encapsulated datagram?

10. What is the destination address of this encapsulated datagram?

11. Is the datagram being forwarded from b to a an IPv4 or IPv6 datagram?

12. What is the source address of this b to a datagram?

13. What is the destination address of this b to a datagram?

14. Is this b to a datagram encapsulating another datagram? Yes or No.

15. What is the source address of this encapsulated datagram?

16. What is the destination address of this encapsulated datagram?

17. Is the datagram being forwarded from a to c an IPv4 or IPv6 datagram?

18. What is the source address of this a to c datagram?

19. What is the destination address of this a to c datagram?

20. Is this a to c datagram encapsulating another datagram? Yes or No.

21. What is the source address of this encapsulated datagram?

22. What is the destination address of this encapsulated datagram?

23. Is the datagram being forwarded from c to B an IPv4 or IPv6 datagram?

24. What is the source address of this c to B datagram?

25. What is the destination address of this c to B datagram?

26. Is this c to B datagram encapsulating another datagram? Yes or No.

27. What is the source address of this encapsulated datagram?

28. What is the destination address of this encapsulated datagram?

29. Is the datagram being forwarded from B to A an IPv4 or IPv6 datagram?

30. What is the source address of this B to A datagram?

31. What is the destination address of this B to A datagram?

32. Is this B to A datagram encapsulating another datagram? Yes or No.

33. What router is the 'tunnel entrance'? Give the router's letter

34. What router is the 'tunnel exit'? Give the router's letter

35. Which protocol encapsulates the other, IPv4 or IPv6?

SOLUTION

1. The datagram is an IPv6 datagram.

2. The source IP address is 46AE:49E8:C9C7:FB83:8DCB:9B88:DCF8:4959

3. The destination IP address is C639:7DAE:85DF:7E3F:5563:557A:4942:AB82

4. No, the datagram is NOT encapsulated.

5. The datagram is an IPv4 datagram.

6. The source IP address is 200.185.208.49

7. The destination IP address is 184.129.134.20

8. Yes, the datagram is encapsulated.

9. The source address of this encapsulated datagram is 46AE:49E8:C9C7:FB83:8DCB:9B88:DCF8:4959

10. The destination address of this encapsulated datagram is C639:7DAE:85DF:7E3F:5563:557A:4942:AB82

11. The datagram is an IPv4 datagram.

12. The source IP address is 200.185.208.49

13. The destination IP address is 184.129.134.20

14. Yes, the datagram is encapsulated.

15. The source address of this encapsulated datagram is 46AE:49E8:C9C7:FB83:8DCB:9B88:DCF8:4959

16. The destination address of this encapsulated datagram is C639:7DAE:85DF:7E3F:5563:557A:4942:AB82

17. The datagram is an IPv4 datagram.

18. The source IP address is 200.185.208.49

19. The destination IP address is 184.129.134.20

20. Yes, the datagram is encapsulated.

21. The source address of this encapsulated datagram is 46AE:49E8:C9C7:FB83:8DCB:9B88:DCF8:4959

22. The destination address of this encapsulated datagram is C639:7DAE:85DF:7E3F:5563:557A:4942:AB82

23. The datagram is an IPv4 datagram.

24. The source IP address is 200.185.208.49

25. The destination IP address is 184.129.134.20

26. Yes, the datagram is encapsulated.

27. The source address of this encapsulated datagram is 46AE:49E8:C9C7:FB83:8DCB:9B88:DCF8:4959

28. The destination address of this encapsulated datagram is C639:7DAE:85DF:7E3F:5563:557A:4942:AB82

29. The datagram is an IPv6 datagram.

30. The source IP address is 46AE:49E8:C9C7:FB83:8DCB:9B88:DCF8:4959

31. The destination IP address is C639:7DAE:85DF:7E3F:5563:557A:4942:AB82

32. No, the datagram is NOT encapsulated.

33. The tunnel entrance is router F

34. The tunnel exit is router B

35. IPv4, in order to maintain compatibility with existing IPv4 infrastructure, IPv6 datagrams are put in the payload of an IPv4 datagram. These IPv4 datagrams are passed on until it reaches a router which supports IPv6, where the IPv6 datagram is decapsulated and passed on.

QUESTION 1 OF 35

Is the datagram being forwarded from D to F an IPv4 or IPv6 datagram?

Answer

CHECK HINT SKIP

We gratefully acknowledge the programming and problem design work of John Broderick (UMass '21), which has really helped to substantially improve this site.

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